

Master in Computer Vision *Barcelona*

Module: C1 - Week 3
Project: Museum painting retrieval
Coordinator: Ramon Morros, Javier Ruiz Hidalgo

W2 - Homework

Query results evaluation

To be updated

W2 - Homework

Query results evaluation

Query results for QST1

Team	Method	map@1
Team8	method1	0.97
Team5	method1	0.97
Team4	method1	0.93
Team5	method2	0.93
Team7	method2	0.93
Team7	method1	0.93
Team1	method2	0.87
Team1	method1	0.83
Team3	method1	0.83
Team6	method1	0.83
Team2	method2	0.77
Team2	method1	0.73

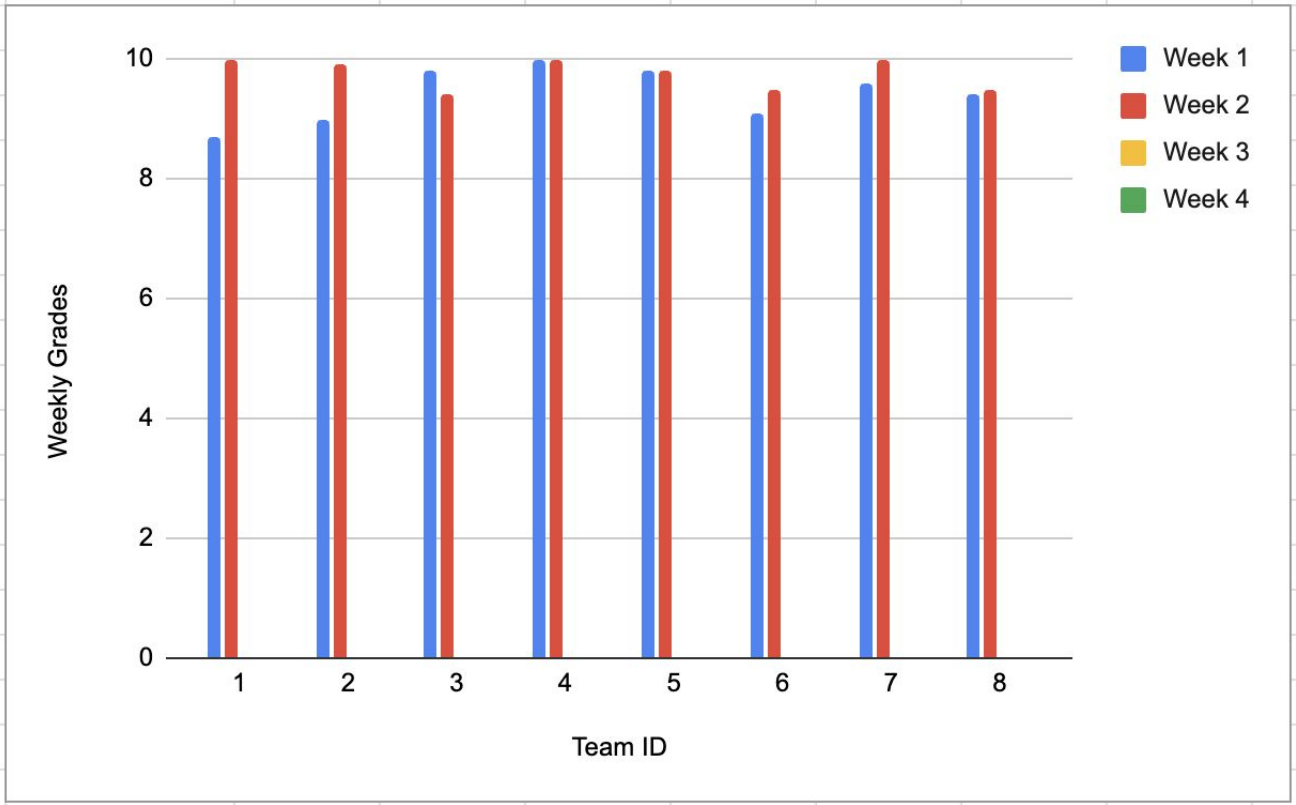
Pixel masks: QST2

Team	Method	Precision	Recall	F1
Team2	method2	0.98	0.97	0.98
Team2	method1	0.98	0.97	0.98
Team8	method1	0.99	0.97	0.98
Team3	method1	0.98	0.99	0.98
Team7	method2	0.98	0.96	0.97
Team7	method1	0.98	0.96	0.97
Team5	method2	0.97	0.97	0.97
Team5	method1	0.97	0.97	0.97
Team1	method1	0.94	0.99	0.96
Team1	method2	0.94	0.99	0.96
Team4	method1	0.97	0.95	0.96
Team6	method1	0.99	0.94	0.96

Query results for QST2

Team	Method	map@1
Team4	method1	0.90
Team7	method1	0.83
Team8	method1	0.83
Team7	method2	0.80
Team3	method1	0.77
Team5	method1	0.77
Team1	method2	0.77
Team5	method2	0.73
Team1	method1	0.70
Team2	method1	0.70
Team6	method1	0.57
Team2	method2	0.53

W2 - Homework grades



W3 - Datasets (II)

Query/Test datasets (development with GT and test without GT)

Paintings with:

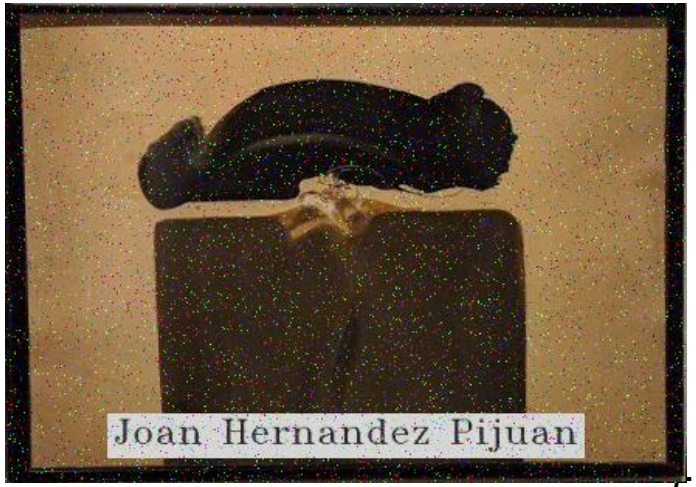
- noise (random samples)
- color changes (random samples, random Hue changes)

Example of color change:



Original Changed

Example of image with noise:



W3 - Datasets (III)

Query Development:

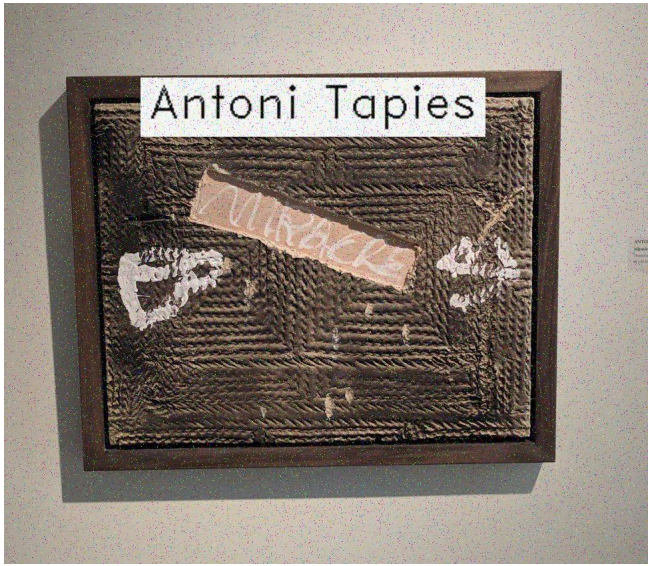
- QSD1-W3 (30 images) cropped pictures, one painting per image, some paintings with noise, some paintings with changes in color
- QSD2-W3 (30 images) non cropped pictures, up to two paintings per image, some paintings with noise, some paintings with changes in color

Query Tests:

- QST1-W3 (50 images) cropped pictures, some paintings with noise, some paintings with changes in color
- QST2-W3 (30 images) pictures with background, in some cases more than one picture per image, some paintings with noise, some paintings with changes in color

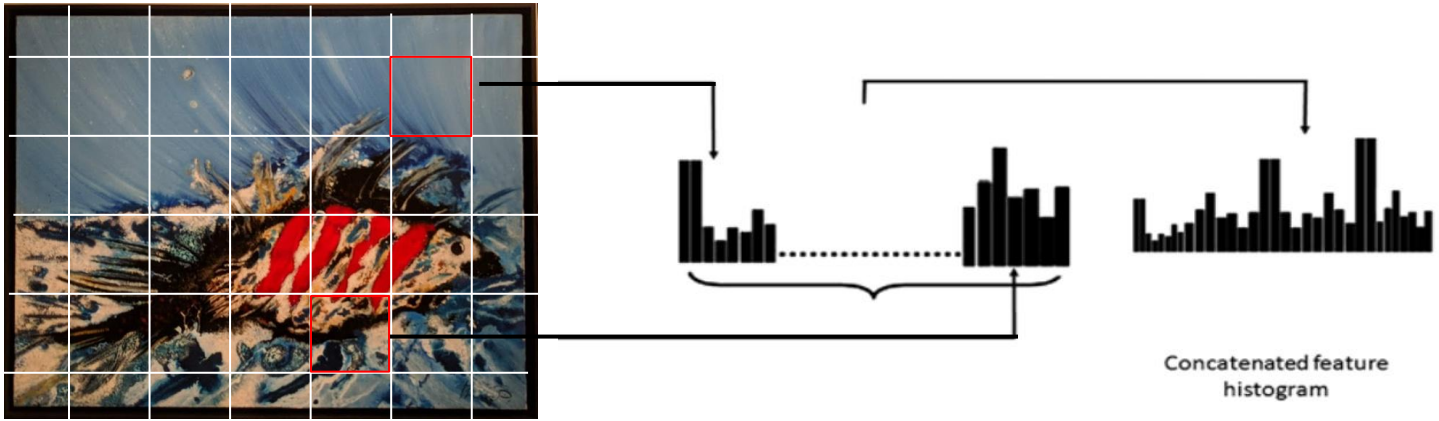
W3 - Task 1

- Filter noise with linear or non-linear filters
  - unknown noise model
  - only some images are noisy
  - show some examples from QSD1-W3



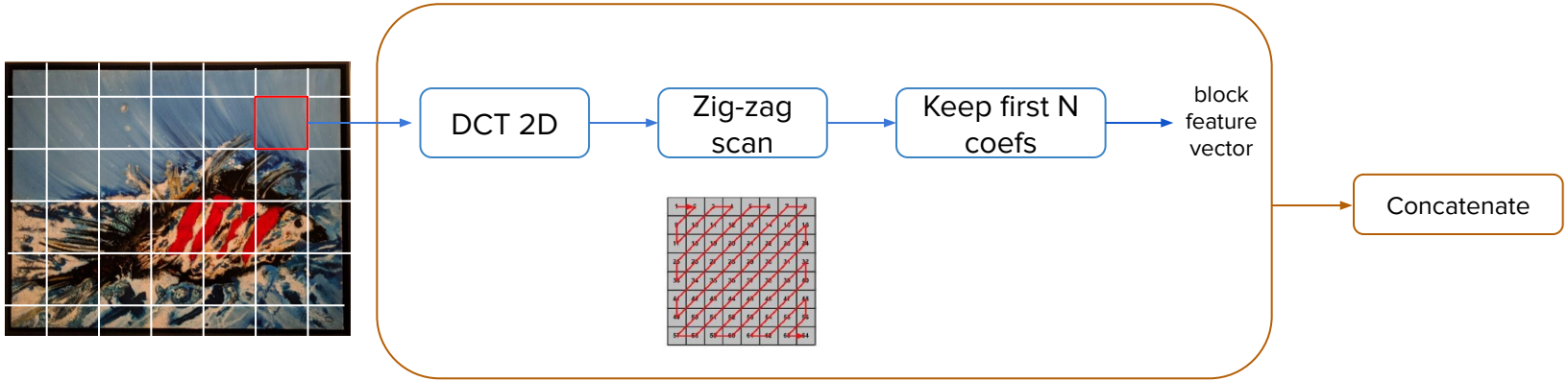
W3 - Task 2 (I)

- Implement texture descriptors (LBP, DCT, wavelet-based, etc.)
- Evaluate query system using QSD1-W3 **using only texture** descriptors
  - LBP, DCT, etc.



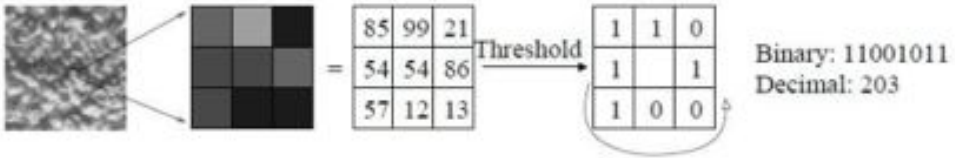
W3 - Task 2 (II)

DCT descriptor:

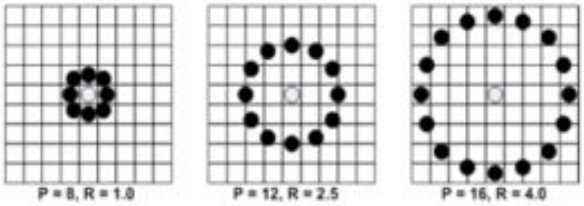


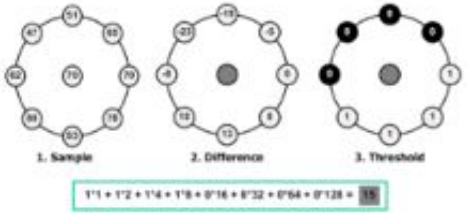
W3 - Task 2 (III)

- LBP (Local Binary Pattern) descriptor
  - Divide image into blocks, for each pixel in the block compare to its 8 neighbors



  - If the center pixel's value is greater than the neighbor's value, write 0, otherwise, write 1. The result is a 8-digit binary number. Compute histogram, over the block, of the frequency of each number
  - **Multiscale LBP:** different local neighborhoods. Bilineal interpolation is performed for points which are not centered on pixel





W3 - Task 3

- For QSD2-W3 : detect all the paintings (max 2 per image), remove background.
  - Create a binary mask to evaluate the method







W3 - Task 4

- For QSD2-W3 : detect all the paintings (max 2 per image), remove background, apply retrieval system, return correspondences.

- Add extra list level to allow for the possibility of more than one painting per image

[[[7,25,...],[44,81,...]], ...

Integers!



[7,25,...] [44,81,...]

W3 - Task 5

Submit your results for a “blind” competition in QST1, QST2

- Add extra list level to allow for the possibility of more than one painting per image
- For QST2, submit the segmentation masks

Note: Deliver pkl files to GDrive folder:
TeamX/week3/QST1/**method1**/result.pkl
TeamX/week3/QST2/**method1**/result.pkl
TeamX/week3/QST2/**method1**/*.png

W3 - Submissions (I)

Deadline slides: Sunday 25 Oct 2026 at 19:00

Deadline results: Sunday 25 Oct 2026 at 19:00

Deadline questions to teams: Monday 26 Oct 2026 at 14:00

Test query set will be published Thursday, 22 Oct 2026 at 14h

Submit progress slides in common google slide template:

- Tasks 1 to 4
- Team questions

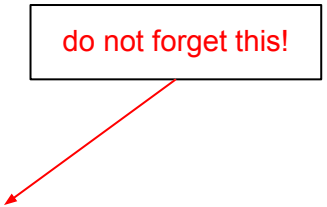
Submit used code in your github repository:

- Team github repository

Submit your test results (next slide)

W3 - Submissions (II)

- For each query test (QST1-W3 (50), QST2-W3 (30)) a list of the K best results (K=10).
  - Only best method (only one for each test set)!



Note: Delive pkl files to:
/home/dlcv0X/m1-results/week3/QST1/**method1**/result.pkl
...
/home/dlcv0X/m1-results/week3/QST2/**method1**/result.pkl
/home/dlcv0X/m1-results/week3/QST2/**method1**/*.png